

SEDAR 4: South Atlantic Snowy Grouper and Tilefish Caribbean Deepwater Snapper Grouper

The SEDAR 4 Data Workshop considered numerous Caribbean and South Atlantic deepwater snapper-grouper species. Data were tabulated for all assigned species, and assessments prepared for the two judged to have adequate data for quantitative assessment – South Atlantic snowy grouper and tilefish (“golden tilefish”).

South Atlantic Snowy Grouper and Tilefish

Tilefish Assessment Workshop

- Ageing discrepancies between laboratories should be resolved. State and Federal investigators should continue efforts to standardize techniques and resolve the systematic discrepancies in age determinations. Additional research should be undertaken to verify and validate age determinations.
- Sampling programs are required to quantify discard rates. Research should also be initiated to identify management strategies that could reduce discard mortality. Discarding may become an increasingly important concern as the stock recovers and compliance with measures such as trip limits become more difficult.
- Fishery-independent data collected by the MARMAP program are important to understanding the dynamics of this population, and the National Research Council has recommended that fishery-independent data play a more important role in stock assessment. However, it has been noted that the MARMAP sampling programs do not have ideal extent, both in area coverage and in sampling intensity, for many important species in the South Atlantic snapper–grouper complex. It would be highly desirable for the MARMAP program to receive sufficient funding to expand its coverage and thus provide improved measures of stock abundance.
- Recent West Coast stock assessments were criticized by the U.S. General Accounting Office (GAO 2004) for not including at least one NMFS (i.e., fishery-independent) data source of sufficient scope and accuracy collected from an unbiased, statistical, and scientifically designed program. Effort should be devoted toward developing an independent data source for the South Atlantic snapper-grouper complex that meets the requirements outlined in the Stock Assessment Improvement Plan and the 1998 National Research Council report on improving stock assessment. This could be done through the MARMAP program or otherwise.

- Representative age, length, and sex composition data are needed for all fisheries, seasons, and areas. Sampling should be distributed according to the pattern of landings. Initial sampling targets are suggested as 20 age structure samples per age and 5 length samples per age sample. This provides approximate tilefish sampling targets of 1000 age structures and 5,000 lengths.
- Additional life history and biological research is needed, especially that which covers the full geographic range of the species. Among other items, fecundity and reproductive research is needed (batch fecundity and frequency at age and/or size).

Snowy Grouper Assessment Workshop

- Ageing discrepancies between laboratories should be resolved. State and Federal investigators should continue efforts to standardize techniques and resolve the systematic discrepancies in age determinations. Additional research should be undertaken to verify and validate age determinations.
- Sampling programs are required to quantify discard rates. Research should also be initiated to identify management strategies that could reduce discard mortality. Discarding may become an increasingly important concern as the stock recovers and compliance with measures such as trip limits become more difficult.
- Fishery-independent data collected by the MARMAP program are important to understanding the dynamics of this population, and the National Research Council has recommended that fishery-independent data play a more important role in stock assessment. However, it has been noted that the MARMAP sampling programs do not have ideal extent, both in area coverage and in sampling intensity, for many important species in the South Atlantic snapper–grouper complex. It would be highly desirable for the MARMAP program to receive sufficient funding to expand its coverage and thus provide improved measures of stock abundance.
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per age sample. This provides approximate snowy grouper sampling targets of 700 age structures and 3500 lengths.

- Additional life history and biological research is needed, especially that which covers the full geographic range of the species. Among other items, fecundity and reproductive research is needed (batch fecundity and frequency at age and/or size).
- Further research is needed into the implications of sex change for fishery management.

Review Workshop

Process and Procedure

- Several members of the Panel found the complete documentation of equations and the inclusion of model code particularly informative, and recommend that such information become a standard component of SEDAR assessment reports. Further, it is recommended that model input data files also be included in future reports.
- The Review Panel suggests that two additional pieces of information be provided in future reports: 1) a table of model parameter estimates, and 2) a thorough documentation of the process that led to the initial model configuration.
- The Review Workshop also recommends that future data workshop reports provide greater evaluation of input data. In many instances data are provided with little consideration of the ‘evaluation of quality and reliability’ as required in the Terms of Reference.
- The Review Panel suggests for future SEDAR's that confusion may be reduced by providing a brief description of the process that leads to assessing only a subset of those species addressed in the Data Workshop.

Research

- Regarding ageing methods, the Review Panel recommends that ageing validation should be accomplished prior to addressing concerns over differences in age determinations between the various labs.
- Regarding age sampling, the Panel recommends that the suggested initial sampling rate for age structures be clarified to avoid the suggestion of age as a sampling strata. The intent is to establish an initial age sample of 20 times the number of ages in the population. The Review Workshop also recommends that stratification by length and development of appropriate age-length keys be considered as a possibly more effective and economical approach to inferring age composition than attempting random age sampling. Regardless of the method ultimately chosen, it is most important to provide

adequate age and length sampling through a rigorous and statistically valid sampling program.

- The Panel recommends exploring the relative importance of age sampling in models of the type used here to assess snowy grouper and tilefish. Such analysis could help identify the best allocation of limited monitoring resources.
- The Panel supports the snowy grouper recommendation regarding research into the implication of sex change. The Review Workshop adds that future assessment models addressing species which undergo sex change should provide model results that incorporate sex-specific information.

Comments of CIE contractors

- The Panel's, and that of subsequent readers', ability to review the Workshop Reports was compromised in that details of analysis and discussion were lost through the multi-step process.
- The acceptance criteria for LFs and AFs could be improved. Acceptance criteria should be based on whether each LF or AF is representative of the catch.
- The way landings were modeled in these assessments could be improved.
- It would be better to estimate selectivities as functions of length, rather than of age.
- Statistical models, like those used here, provide a powerful tool for dealing with uncertainty. They allow us to assign appropriate weights to different sources of information and they tell us how certain we can be about our inferences. In practice it is impossible to gain the full power of these models because we are unable to correctly specify all the statistical components of the model and so are often forced to add arbitrary non-statistical components. I suggest that our aim should be to minimize these non-statistical components.
- There is clearly a need for validation of the ageing of both species so that we can have more confidence in the AFs and the age-length conversion matrix.
- The MCB analyses are a good way to replace one type of sensitivity analysis whose aim is to quantify uncertainty. Another type of sensitivity analysis which could have been useful in the Workshop would have been to rerun the initial run several times, each time dropping one type of data, thus showing the extent to which the assessments depended on each data type.
- There were several small problems in both assessments, mostly in the documentation.

- It should be made clear that the calculation of generation time involves only female fish
- In fitting the von Bertalanffy equation the assumption used was clearly that the standard deviation of length at age was proportional to the mean length (not the variance, as stated).
- In the formula for the age-length conversion matrix the superscript 2 is misplaced.
- Equations should be given for the per-recruit calculations.
- It might be worth checking the method of fitting the maturity ogives for both species because the fitted curve is to the right of all data points for which the proportion mature is not near 0 or 1
- In the tables documenting the model it might avoid confusion if a clear distinction were made between fixed parameters (e.g., growth parameters, LF sample sizes), estimated parameters (e.g., selectivity parameters, fishing mortalities), derived quantities (e.g., length at age, selectivity at age) and observations (which are characterized by having an associated likelihood component, e.g., CPUE, LFs).
- Snowy Grouper : It might be useful to try some more sophisticated techniques (e.g., GAMs or tree-based regression) to seek an explanation of the unrealistic MCB runs. This may be informative. It might be worth dropping the Chevron trap CPUE index (for reasons given above). It seems a matter of some concern that more than half the catch is of immature fish. It is worth considering explicitly modeling the three categories of fish: immature, mature female, mature male (i.e., keeping track of numbers of fish by age and category)
- Tilefish: It would be worthwhile to explicitly model sex (i.e., to keep track of numbers by sex, as well as by age — the assessment report stated that this was not possible because the landings and LFs were not sex-specific, but I don't see why). As females are smaller at age than males they probably do not have the same selectivity at age as males do, so modeling selectivity as length-based would be better.

Caribbean Deepwater Snapper Grouper

Data were compiled for several Caribbean Deepwater snapper grouper species during the SEDAR 4 data workshop. Significant data deficiencies were noted, leading to an extensive list of recommendations.

Landings Statistics

Puerto Rico

- In Puerto Rico it is important to determine the feasibility of expansion factors to estimate total catch. The information used to calculate expansion factors by year needs to be verified. Reporting of single trips, rather than multiple-trips per record in the catch report forms should be encouraged. This would greatly facilitate the estimation of effort and CPUE.

U.S. Virgin Islands

- The collection of landings statistics in the U.S.V.I. should also aim at breaking down the reported catch into species, since analysis of the current species-groupings is not straightforward without additional information on species composition from TIP or alternative sampling programs.
- The information used to calculate expansion factors by year (number of fishermen registered, reporting, etc.) needs to be corroborated, and the feasibility of these expansion factors for estimation of total catch needs to be determined.
- Further examination and analysis of the data sets available to date would require an improved collaboration between local and SEFSC biologists. In particular, it is important to determine what species were commonly grouped within each gear-type classification in the 'Old Report Forms'. This information would help to break up the aggregated catch from years prior to the implementation of the Trip Interview Program.
- Landings files for most years for the period 1974-2002 have now been compiled and provided to the SEFSC. However, some coding, typing and other errors, duplicates, as well as gaps in the time series still persist. Action is required to verify, correct the errors and edit those data for future use.
- Significant effort should be geared toward the standardization of the landings series.
- Finally, it would be important to encourage fishermen to submit all the monthly catch reports, to submit reports for months when they do not fish, and to complete all the fields in the reports, since critical information such as effort, gear, and location fished are often missing or incomplete.

Trip Interview Program (TIP)

General Recommendations

- Encourage the development of length-weight equations from the existing information in TIP

Puerto Rico TIP

- Record the total weight landed by species for each trip.

- Record the sampled fractions.
- Coding errors in length and weight units must be corrected.

U.S. Virgin Islands TIP

Encourage/ aid the development of a commercial logbook system to enable estimation of reporting fractions.

- Increasing the fraction of interviewed trips (the sampling fraction needed to achieve specific objectives will depend on the objective and the variability of the observed species composition) to properly determine the species composition, which is needed to break out the aggregated catches.
- Conduct regular interviews in St. Thomas and St. John, with the goal of increasing the sampling fraction.
- Encourage port samplers to complete all the fields in the sampling form. Often the trip effort information is missing, which is essential for the estimation of catch rates or relative indices of abundance.
- Continued and enhanced collaboration between the NMFS SEFSC scientists and the local USVI biologists and data collection agents.
- Correct coding errors, particularly in length and weight units.
- Some questions that could be posed to local USVI biologists to improve the analyses of TIP data are:
 - * How is the species in question landed, gutted or whole, etc.? How are length and weight typically recorded?
 - * Is the species in question targeted or by-catch of another target species?
 - * What species are often landed in association with a given species?
 - * Is the species ever reported under a different name? For example, another species id, or a genus or family designation?
 - * Are there environmental factors that might influence the abundance or catch rates of a given species?
 - * Have management efforts, economic impacts, weather events, or other factors influenced fishing effort, catch rates or targeting?

- * Have fishery attributes changed (gear, boat type, technology, species targeted, skill of fishers etc.) changed during the period of monitoring. If so how?
- * Are interviewed trips chosen randomly? If not, what potential biases might exist in the dataset?

Catch Rates

- In Puerto Rico the total catch by species for each trip in the TIP data is required. It has to be determined whether assumptions can be made regarding sampling fractions in TIP data to allow construction of Puerto Rico's CPUEs.
- The SEDAR Committee recommended that CPUEs for the U.S.V.I. be recalculated for a truncated time series (1984-1991), given that sample sizes for subsequent years are very limited.
- It is important to explore the availability of other fisheries-independent CPUE series.
- Standardization approaches for data-poor species, different from the delta-lognormal, need to be evaluated.
- The use of bootstrapping to estimate confidence intervals of the CPUE series should be investigated.
- The use of multivariate statistical analysis is recommended to identify the appropriate pool of gears to use when measuring effort.

Species Composition

- In Puerto Rico, it is important to recommend increased interviews with an emphasis on representative sampling, and to record the sampling fraction.
- Eventually, if Puerto Rico moves toward reporting landings by species, it will be advisable to compare TIP and landings species composition.
- In the U.S.V.I., it is important to examine the species composition on handline and trolling trips separately, and to evaluate whether sampling is representative.

SEAMAP Survey

- Encourage continued annual surveys throughout the area.
- Determine the spatial/temporal coverage in fine detail.
- Data analysis and interpretation must address the temporal patterns observed in the size frequency distributions.

- Regarding the shallow reef fish monitoring fishery-independent survey in Puerto Rico:
- * Coordinate with NMFS to make this data readily available.
- * Explore the CPUE and size-frequency data available from this data set.
- * Compare with the other SEAMAP data set.

General Recommendations

- Continue and improve collaboration with scientists from Puerto Rico and the U.S. Virgin Islands. Advice is needed in terms of handling the data, interpreting it, correcting coding errors, duplicates, and other problems in the data collection, recording, and editing systems. Local scientists and staff can help to understand the sampling protocols, documenting the observed trends, and filling out persisting gaps in the time-series.
- Continued data exploration must be made with consultation of the local laboratories/agencies, including the biologists, field agents, and data-entry staff.
- There is a possibility that the data will have limited value for assessment in the near future; however, continued analysis and improved data collection may greatly increase the utility of the information. The fishery-dependent data from Puerto Rico in particular has a good potential for use in stock assessment.
- Emphasis should be placed on the improvement of the TIP sampling program, as catch rate standardization, catch composition and size-frequency analyses will continue to rely upon this information. However, fishery-independent surveys and the collection of other biological data are extremely important to develop alternative indices of abundance.
- It is recommended that early biological or biostatistical sample data for the U.S. V.I., from the early to mid 1970's be computerized and made available for future data workshops. It is strongly recommended that formal discussions between NMFS, SEFSC TIP program coordinator and the USVI DFW are held to ascertain what steps/procedures, etc. are needed to improve sampling in the U.S.V.I. fisheries. Similarly, discussions should be initiated between Puerto Rican biologists and NMFS assessment staff to identify any remaining historical data sets not yet available. It is noted that an effort to computerize Puerto Rico biostatistical samples from the mid 1980's is ongoing (N. Cummings personal communication).
- It is recommended that analytical efforts expended by the recent working group members be continued. First, some attention should be given towards identifying or selecting which species should be assessed more quantitatively. The Caribbean reef fish fisheries are complicated comprising a mix of many species that are harvested by a number of gears.

- It is recommended that additional workshops such as this one be implemented to further develop the information for assessment, especially for those species and fisheries for which extensive information exist.
- It is noted that that strong cooperation of all agencies and local scientists involved would be beneficial.

Availability of Data for Stock Assessments

The workshop participants reviewed summaries of the information presented by the Caribbean group which might be used to assess the status of silk, queen and blackfin snapper and sand tilefish on each platform (Table 64). For the Puerto Rican platform the availability of information was examined for three data sources: Puerto Rico, the United States Virgin Islands and the British Virgin Islands.

Puerto Rican platform

- For Puerto Rico, reported commercial landings are available in electronic form only since 1983 although the local fisheries were exploited since the early 1900's. Efforts are underway to obtain previously computerized data files of landings for 1963-1982 (N. Cummings personal communication). These early landings statistics could better characterize fishing mortality levels on this multi-gear/multi-species fisheries complex and efforts should be made to extract these data. Snapper landings in the Puerto Rico database are apparently aggregated for multiple species within the 'silk snapper group' in the Puerto Rico database before 1997 (after 1996 silk snapper is apparently not aggregated with other species) (Aida Rosario personal communication). Estimates of the landings of those snappers probably can be made given some assumptions about the species composition information from dockside sampling after considerable additional effort and consultation with Puerto Rican biologists who are familiar with the data collections and fisheries. It is strongly recommended that cooperative analyses be initiated between scientists from Puerto Rico and NMFS to accurately quantify species composition from these data. Analyses should take into account the highly variable operations of the local fisheries.
- For sand tilefish annual landings are less than 1,000 lb and in most years less than 50 lb. The dockside sampling (TIP) data (which might be used for species composition had very few sand tilefish recorded) so that if sand tilefish landings had been included in the various unclassified categories, it would not be possible to estimate the amount of sand tilefish in such landings.
- For Puerto Rico the recreational harvest of the three snapper species are thought to be relatively low compared to the commercial landings. Because they are thought to be low, the absence of recreational harvest estimates prior to 2000, was thought not to be a major problem for assessment of these stocks, given the other uncertainties in the data sets.

- The landings in the United States Virgin Islands have not been recorded by species; therefore species composition information would be needed for St. John and St. Thomas to estimate catches by species. Only limited species composition samples have been collected from those islands, therefore estimates of the landings by species have not been made. Additionally there is no information on possible recreational harvests of these species around those islands. Also there was no information available at the workshop on the British Virgin Island fisheries. It is noted that an effort is ongoing to obtain historical information on landings and biostatistics samples for the British Virgin Island (BVI) fisheries for use in future data workshops regarding the Puerto Rico platform. It is also recommended that biologists from the BVI fisheries department be included in future data-workshops that involve the appropriateness of the use of data from the BVI in characterizing reef fish stocks on the Puerto Rican platform.
- Information on size composition is available for the three snappers from the Puerto Rican commercial fishery and a limited amount of information is available for silk snapper from the recreational fishery. Additionally, there are ample observations on the size of sand tilefish taken in the fishery-independent sampling near Puerto Rico, and there possibly are sufficient samples for silk and blackfin snappers from those surveys. For St. Thomas and St. John there are few or no size samples from the commercial and recreational fisheries. The workshop participants have not determined whether there were fishery independent samples from that area.
- It is expected that crude information on commercial catch rates could be obtained for the three snappers from expanded annual landings and estimated deep water effort for Puerto Rico; it seems unlikely however that the TIP data could provide reliable indices of abundance for those species, because it does not appear that the total landed weight for a species was recorded and it appears that in general not all fish were measured. It is likely that the fishery independent sampling could be used to develop an index of abundance for sand tilefish, and probably also for silk and blackfin snapper. There do not seem to be sufficient data for calculating fishery dependent catch rates from St. John and St. Thomas.
- In summary for the Puerto Rican platform:
 - * For the Puerto Rican platform it seems that multiple years of commercial landings might be developed for the three snapper species from reported catches and species composition data. However it would best if these tasks were done in consultation with scientists familiar with the fisheries and the specific datasets. Those catches would however represent only a part of the total removals.
 - * Some information can probably be obtained from the TIP collected size frequency of the commercial catch for the three snappers. It is recommended that analytical

effort focus on further review of the available size frequency samples. Of all of the available data sets, the fishery independent sampling on the Puerto Rican platform conducted by the NMFS, SEFSC and by the PR, DNER, FRL FSP may be most likely to provide indications of the abundance trends of at least silk and blackfin snapper and sand tilefish on the Puerto Rican platform. It is recommended that analytical efforts focus on aggregating those data sets and developing abundance indices.

St. Croix platform

- The landings data from St. Croix probably can be disaggregated into species-specific data sets, but is restricted to a limited number of years when species composition is available and the landings are categorized by species-groups. However, the generally low sampling fractions indicate that there would be very great uncertainty about the estimated landings by species. Disaggregating the catch from the earlier years, when no species composition is available and landings were recorded by gear category may be cumbersome. Added to these issues is the possible imprecision in the estimation of the total catch based on expansion factors. These will be more reliable once compliance reports are reviewed and reanalyzed for the full time-series. Given these uncertainties, the overall utility of the catch for use in stock assessment is questionable at the moment, particularly for years prior to the implementation of the TIP program.
- The decrease in the mean size and the size of the larger (80th percentile) of both silk and queen snapper landed in St Croix between 1983-1996 could have been an indication of over harvesting. Additionally that the majority of silk snapper are below the estimated size at maturity would have been of substantial concern if fishing mortality rates were high. The standardized commercial catch rates calculated from the TIP samples from St. Croix were based on relatively few observations and the time series ends in 1991 (too few observations in subsequent years). Thus, they do not provide information on the current status of the resource. It is recommended that cooperative efforts be initiated by NMFS, SEFSC and the USVI DFW to address improvements in sampling the near-shore reef fish fisheries off these islands. It is possible that the fishery independent sampling (1992-1994, 1999, 2002) conducted by the NMFS, SEFSC Pascagoula Laboratory off the Virgin Islands could provide useful information, but it was not clear to the workshop participants what portion of that sampling occurred on the St. Croix platform. Once again it is recommended that examination of the fishery independent data be given high priority in terms of expending analytical time.
- Participants at the workshop understood that additional fishery independent data sets may exist for both the Puerto Rican and the St. Croix platforms particularly from in situ observations. It was recommended that efforts be made to obtain that information for possible use in developing additional indications of population status.